

SEQUENTIAL CANDIDATE SELECTION FOR SCORE FOLLOWING IN SHEET MUSIC IMAGES

Pouya Mohseni, Ichiro Fujinaga

Schulich School of Music, McGill University, Montréal, Canada

ABSTRACT

Score following aligns a live performance to its score in real time, committing to a position at every instant with no lookahead. The two sides are different kinds of object, the performance audio and the score an image, and the established response is to convert one into the other and align within a single modality, by optical music recognition or by transcription, each detour inheriting the errors of its conversion. Sheet-image score following converts neither side and learns the correspondence directly, conditioning a visual detector on the audio. What limits it is annotation, since performances aligned to notehead coordinates are costly enough that nearly all training data is synthesized, so these systems work on synthesized audio and lose much of their accuracy on a real instrument in a real room. Every remedy proposed for that gap has treated it as a failure of perception. We present evidence that it is not, since on real recordings the detector already proposes a position within the standard tolerance at every scored onset and an oracle held to the same realtime commitments recovers almost everything the deployed rule loses. We therefore recast the task as sequential selection over the candidates the detector already emits, combining a parameter-free transition model with a set-based scorer that judges each candidate against its competitors in the same instant. This sets a new best on real piano recordings played straight through, holds across three released detectors, and shows that what the decision needs is its own temporal history rather than a better image representation.

Index Terms— Score following, sheet music, audio-to-score alignment, realtime, music information retrieval

1. INTRODUCTION

Music alignment establishes the correspondence between a performance and the score it realises, mapping each moment of the audio to a position in the written music. It divides on whether the alignment is allowed to look ahead. Offline, where the task is called audio-to-score alignment or music synchronisation, the whole recording is available, a global method such as dynamic time warping settles every position at once [6] and even repeats and jumps can be resolved after

the fact [7, 8], which is what underpins cross-modal retrieval and piece identification [9]. Online, the system observes only what has been played, commits to a position immediately and never revises it.

That online case is score following [1], also called music tracking [10]. Given the audio observed up to frame t it estimates the position the performer currently occupies, and must do so before frame $t + 1$ arrives. It supports automatic page turning [2], live computer accompaniment [3, 4, 5] and score-synchronised visualisation. Everything below respects that constraint, the oracles included.

Its difficulty is that the two sides are different kinds of object. The performance is audio and the score an image, with no shared space in which to compare them, a mismatch the literature answers in three ways.

1.1. Related work

Two families make the sides commensurable by converting one of them. Converting the score, optical music recognition turns the page into symbols so that a classical follower can track over a discrete set of states, a well-developed line spanning online dynamic time warping [6], duration-focused probabilistic models [11], hidden semi-Markov models absorbing performer errors and arbitrary repeats [12], particle filters for rests and tempo change [13], switching state-space models with hidden tempo [14] and multi-agent tracking [10]; its weakness is that optical music recognition remains error-prone on real scores [15] and the follower inherits every mistake. Converting the audio instead, recent work pairs realtime piano transcription with symbol-level tracking to strong effect [16], though the follower is then bounded by the transcriber and restricted to instruments transcription handles well.

The third family converts neither side and learns the correspondence between audio and pixels directly. Dorfer et al. [17] classified position into staff-level buckets on crops; reinforcement learning framed tracking as an agent navigating a score strip [18, 19]; CUNet [20] moved to full pages with an audio-conditioned segmentation whose centre of mass gives the position; and CYOLO [21] recast it as audio-conditioned object detection, CYOLO-SB [22] adding heads for the enclosing system and bar. The concurrent CODA [23] cascades system, bar and note selection over layout regions with beam

search and learned transition penalties, and is the strongest published system on this benchmark, though it reads the system and bar boxes of the page at test time, which the other methods and ours do not.

All of them need performances aligned to notehead coordinates, an annotation costly enough that nearly all of it is synthesized, a score rendered to MIDI and the MIDI to audio. The result is systems that work on synthesized audio and lose much of their accuracy on a real instrument in a real room, CUNet [20] falling from 85.5% of onsets tracked within 0.5 s to 22.4%, CYOLO-SB [22] from 89.3% to 79.9% and CODA [23] from 95.5% to 88.3%. Closing that gap is the field’s central open problem, and every remedy proposed for it has been a perception remedy, whether reverberant augmentation [21], extra proprietary recordings [22] or a new architecture [23].

1.2. Diagnosis

We tested that premise before accepting it. A conditional detector does not produce a position directly but scores thousands of candidate noteheads and reports the most confident. Keeping the 256 most confident, on real room-microphone recordings a candidate within the 0.5 s tolerance is present at every one of 4,149 scored onsets, so an oracle free to pick any of them is perfect but could not be run live. The oracle that matters commits at each instant and never moves backwards, exactly as a deployed follower must, and still reaches 99.6% where the deployed rule reaches 80.0% (Fig. 1). Nineteen of the twenty points lost to real audio survive the realtime constraint. At this tolerance, then, reverberation degrades the detector’s confidence ordering more than the visual evidence beneath it, and Sec. 4 qualifies how far that holds as the tolerance tightens.

We therefore recast the task as a sequential decision over the positions the page affords rather than the regression of a coordinate. CANDOR decides with a transition model scoring a displacement against elapsed time and a set-based scorer judging each candidate against its competitors in the same instant, trained listwise on synthesized audio alone, causally and greedily, at 1.25 ms per instant on one CPU thread against the 50 ms available at 20 frames per second.

Our contribution is to relocate the synthetic-to-real gap in this task and to act on it. Candidate-set oracles place it in the decision rather than in perception, and CANDOR, a decision model built on that finding, sets a new best on real recordings by a margin holding across three released detectors differing in heads and in training data.

2. METHOD

2.1. Backbone

Every sheet-image follower of the last five years shares one perception module, a convolutional network over the score page whose feature maps are modulated by the audio. The

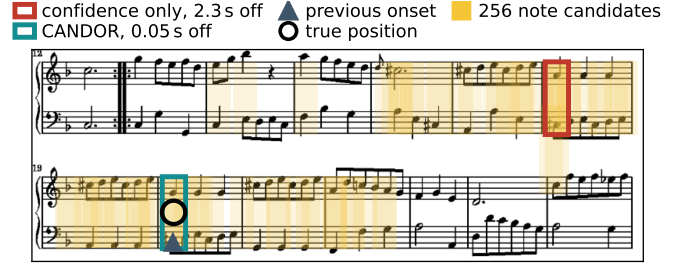


Fig. 1. One instant of a real room-microphone recording. In yellow, all 256 candidates the detector scores, drawn at one opacity so that where they pile up the fill darkens. The most confident (red) lies on the wrong staff, 2.3 s from the truth; CANDOR (teal) takes a less confident candidate on the true note (ring), forward of the previous decision (caret). The correct position was available and was not chosen, which is the characteristic failure on real audio.

mechanism is feature-wise linear modulation [24], which injects an external vector into a network by scaling and shifting its activations,

$$f_{\text{FiLM}}(x) = s(z) \odot x + t(z), \quad (1)$$

where s and t are learned linear maps of a conditioning vector z and $\odot$ is elementwise. The page alone cannot say where the performer is, so z carries the recent audio and the visual features become a function of what has just been heard.

CYOLO [21] encodes the last two seconds of audio into z with a spectrogram network and places FiLM layers through the downscale and upscale blocks of a YOLO detector [25], which then predicts a bounding box for the notehead currently sounding. CYOLO-SB [22] keeps that arrangement and adds heads for the enclosing system and bar, so one 52×52 feature map yields all three. CODA [23] conditions its own backbone the same way, at the deeper stages, before cascading selection over layout regions supplied with the score.

We take this module as given rather than as a design choice, and use the released CYOLO detectors to propose noteheads. No detector weight is updated anywhere in this work, and three released instances are used, differing in their heads and in their training data, so the conclusions concern the family and not one checkpoint.

2.2. CANDOR

CANDOR, for candidate ordering, sits after the backbone and replaces only its final step. At each instant the detector offers a few hundred possible notehead positions, and where it would report the most confident of them, CANDOR asks which one is most compatible with where its own previous decision was and with what the rest of the set looks like. Those two questions are the two parts below, and answering them requires carrying state from one instant to the next.

2.2.1. Candidate positions

A performance is processed in instants $t = 1, \dots, T$, and at each one the backbone reads the audio so far and the current page and scores note boxes $b_i = (x_i, y_i, w_i, h_i)$ with confidence o_i ; we keep the $K = 256$ most confident, $\mathcal{C}_t$. Staves are concatenated along one horizontal axis, the unrolled coordinate u_i , so positions across a line break are adjacent. The released detectors score 8,112 boxes per instant and report

$$\hat{u}_t = \arg \max_{i \in \mathcal{C}_t} o_i, \quad (2)$$

which is the degenerate selector. CANDOR reads $\mathcal{C}_t$ and its own previous choice $\hat{u}_{t-1}$, and nothing else. K is set where availability stops binding, since a correct candidate is present at every onset only from 256 and the causal oracle falls to 85.6% at $K = 64$, below what CANDOR itself reaches, so a smaller set would limit the decision rather than test it.

2.2.2. Transition model

Selections must form a plausible trajectory, and between instants Δ_t frames apart the performer advances by a roughly predictable amount, so the displacement $d_i = u_i - \hat{u}_{t-1}$ is scored by a floored Gaussian whose mean grows with elapsed time,

$$\log \pi_i = \max \left(-\frac{(d_i - \mu \rho_t)^2}{2\sigma^2}, J \right), \quad (3)$$

with $\rho_t = \text{clip}(\Delta_t/\Delta_0, 0.2, 8)$ and $\Delta_0 = 5$ frames, the mean inter-onset gap in training. The floor J keeps a jump possible at bounded cost. Combined with confidence, $h_i = \log o_i + \log \pi_i$ already recovers a large part of the gap on its own.

2.2.3. Set-based scorer

The transition model treats candidates independently given the history, so what it cannot express is comparative, whether this box, among those on offer in this instant, looks like a real notehead rather than a spurious response. This is what real audio in particular calls for, since reverberation smears the onset the conditioning vector is built from and the detector answers with many moderately confident boxes rather than one sharp response, so absolute confidence is poorly calibrated while the shape of the set is still informative. Each candidate is described by $\phi_i \in \mathbb{R}^{33}$ holding its confidence relative to the instant's best, box geometry, motion relative to the history, relation to any coarse regions the backbone emits, and a neighbourhood group describing how crowded and how typical the box is among its neighbours. We also read f_i , the 128-dimensional FiLM-conditioned feature from which the detection head scored the box, so the decision sees the representation perception used. A permutation-invariant network [26], as in learned re-scoring of detections [27], embeds the set and pools a context shared

by all candidates,

$$e_i = g([\phi_i; \text{ReLU}(W f_i)]), \quad c = \left[\frac{1}{K} \sum_j e_j; \max_j e_j \right], \quad (4)$$

$$s_i = r([e_i; c]), \quad (5)$$

with g and r two-layer perceptrons (64, 32 and 64, 1) and $W \in \mathbb{R}^{64 \times 128}$. The maximum is taken per dimension, so every candidate is judged against its competitors.

2.2.4. Training and inference

The decision is supervised listwise, so with ε_i the error in frames of candidate i the target is $q_i \propto \exp(-\varepsilon_i/\tau)$, $\tau = 3$, and

$$\mathcal{L} = - \sum_{i \in \mathcal{C}_t} q_i \log \frac{\exp s_i}{\sum_j \exp s_j}, \quad (6)$$

so the scorer learns to order a set rather than to regress a coordinate. History features come from previous decisions, perturbed on 30% of samples by Laplace noise of scale 30 px to expose training to the drift of sequence prediction [28]. At inference CANDOR decides

$$\hat{u}_t = \arg \max_{i \in \mathcal{C}_t} \beta s_i + (1 - \beta) h_i, \quad (7)$$

causally and greedily, one pass over 256 candidates per instant, in 1.25 ms on a single CPU thread against the 50 ms available at 20 fps. Training on the residual $s_i - h_i$ and deciding $\arg \max_i (s_i + h_i)$ is Eq. 7 at $\beta = 0.5$ and eliminates all four constants.

3. EXPERIMENTAL SETUP

3.1. Dataset

All experiments use MSMD [9], which pairs public-domain LilyPond engravings with MIDI-derived alignments for solo piano. Its repertoire is Western classical, weighted towards the Baroque and Classical periods and dominated by Bach, with Beethoven, Mozart, Handel, Schumann, Chopin, Czerny and Burgmüller making up most of the remainder. Textures range from two-voice inventions to dense Romantic writing, which matters here because the number of simultaneous noteheads sets how many plausible candidates a page affords at each instant.

The decision model is trained on 353 pieces, synthesized and convolved with measured room impulse responses [29], and validated on 19, using AdamW ($3 \cdot 10^{-3}$, cosine, 40 epochs, batch 256). Evaluation is the real-performance benchmark of [19], 16 pieces of Bach, Schumann, Mozart, Chopin and Mussorgsky spanning 25 pages and 4,149 onsets, played on a Yamaha AvantGrand N2 and captured with a room microphone. A further 80 held-out pieces are rendered with a real impulse response and noise from 12 to 0.5 dB SNR. We report the percentage of onsets whose predicted position maps to within

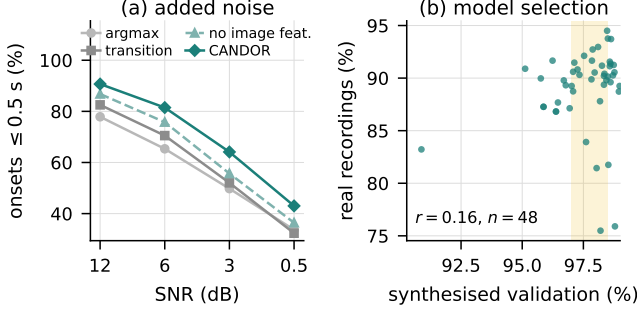


Fig. 2. (a) 80 held-out pieces under a measured room impulse response and additive noise, 40,594 onsets per level. (b) Each point is one of 48 trained decision models. Half of them fall in the shaded 1.5-point band of validation accuracy and still differ by 19 points on the real recordings.

0.05 to 5 s of the true time, 0.5 s primary, with 95% intervals bootstrapped over pieces [30], and seed means with deviations.

Three kinds of data enter at three stages and are worth keeping apart. The decision model is trained on synthesized audio alone and never sees a real recording. The four constants (μ, σ, J, β) are calibrated on real recordings leave-one-piece-out, chosen for each evaluation piece on the other 15 with pieces weighted equally. Evaluation is on the piece held out. Leaving a piece out of its own constants does not stop the benchmark as a whole from participating in choosing them, which the published baselines did not require, so the residual form of Eq. 7 is carried alongside as the variant that fits nothing to any real recording.

4. RESULTS

CANDOR orders proposals, so it applies to a follower that produces a candidate set and then discards it. CODA [23] is not such a target, since its final stage regresses a continuous bar-local coordinate rather than choosing among proposals, and the sets it does select over, systems and bars, are supplied by layout annotation and already decoded with beam search. It appears in Table 1 as the strongest published system on these recordings, not as a backbone CANDOR could be attached to.

Table 1 sets CANDOR against the published literature, taken from [23], on recordings played straight through; Sec. 4.2 reports what happens when a performance is not, and Δ is the gain at 0.5 s over the perception model each row is built on. CANDOR tracks 94.4% of onsets within 0.5 s, 95% CI over pieces [92.2, 97.1], against a previous best of 88.3%, and leads at every threshold. Sixteen pieces are few and the interval is correspondingly wide, though the gain over the argmax it replaces, paired on the same resample, is +7.9 with interval [+2.8, +12.2]. Nor is the margin an artifact of one perception model. Retrained on whichever candidates each emits, the decision gains 20.4, 10.7 and 7.9 points, and on

the weakest, whose training data is entirely public, it passes where the strongest one begins. A decision model does not need good proposals, only proposals containing the answer. On 94 synthetic test pieces it reaches 95.5% against 89.3% for argmax, and on the 80 noisy held-out pieces it beats the transition model at every level from 12 to 0.5 dB (Fig. 2a).

The tightest thresholds qualify this. On CYOLO-SB candidates CANDOR reaches 69.0% and 71.4% at 0.05 and 0.1 s, below CODA’s 72.1% and 74.3%, since a decision is only as precise as the boxes it chooses between and the mechanism recovers which notehead is active rather than the position within it.

Table 2 separates the parts over the same candidates, adding Eq. 3 to Eq. 2 and then the scorer, with n counting decision-model seeds at a fixed detector. The transition model alone, with no learned parameters, accounts for roughly half the gain. The residual variant is the one to read for the strictest protocol, fitting no constant to any real recording and so comparable with the published rows on equal terms; it reaches 88.7% against the 80.0% it replaces, above the previous best, and the distance from there to 94.4% is what calibrating four constants is worth.

4.1. Ablation study

Giving the scorer the FiLM-conditioned backbone feature, DINOv2 patch features [31] on the same grid, or no image feature at all yields 90.6, 90.9 and 90.9, a spread well inside the 0.9-point seed deviation, where on synthesized validation audio the same three differ by 3.4 points. Those features are learnable and useful there and worth nothing on real audio. Removing the features computed against the previous decision instead costs 7.2 points, the largest single ablation here, and teacher forcing attributes two thirds of the remaining error to drift, since given the true previous position the same scorer reaches 98.0% against the causal oracle’s 99.6%.

We also rebuilt the perception model, retraining the detector from scratch for a fixed 24-hour budget with a different audio encoder. These are the lower block of Table 1, not comparable with the converged checkpoints above it and reported because the decision gains a similar margin whatever it is attached to. A frozen MERT-95M tower [32] with only a projection trained reaches 46.2 and overfits 12.6-fold. Replacing the whole audio tower with the two-layer causal Mamba [33] of [23], trained as they train it, reaches 76.2 against 83.3 for an identically budgeted control, while keeping the convolutional front end and replacing only the recurrence, which we call CYOLO-Mamba, reaches 91.5. The recurrence therefore benefits from a state-space model and the convolutional front end is load-bearing, which is the encoder ablation absent from [23]. Replacing the visual stem with frozen DINOv2-base patches on the same 52×52 grid, against a control retrained identically, gives .. None of this rescues the gap on its own, the strongest of these detectors merely reaching what re-deciding over an

Table 1. Results comparison on the real room-microphone recordings. † and ‡ mark proprietary training data and a layout annotation read at test time.

Method	Onsets tracked (%) within threshold (s)					Δ
	≤ 0.05	≤ 0.1	≤ 0.5	≤ 1.0	≤ 5.0	
MM-Loc [17]	36.4	40.6	58.5	61.1	73.5	
RL agent [19]	18.5	20.0	47.6	60.3	90.1	
CUNet [20]	11.3	12.5	22.4	26.6	44.3	
CYOLO [21]	56.3	58.1	71.2	74.9	91.9	
CYOLO-SB [22]	61.0	63.0	79.9	83.2	96.0	
CYOLO-SB+A [†] [22]	68.2	70.6	86.5	89.1	98.1	
CODA [‡] [23]	72.1	74.3	88.3	91.4	98.7	
CANDOR _{CYOLO}	72.6	74.7	91.6	95.0	99.1	+20.4
CANDOR _{CYOLO-SB}	69.0	71.4	90.6	94.2	98.2	+10.7
CANDOR _{CYOLO-SB+A[†]}	76.0	78.5	94.4	96.9	99.6	+7.9
<i>Detectors we trained, 24 h from random initialisation; only the audio tower differs</i>						
CANDOR _{CYOLO-LSTM} (control)	58.2	60.6	83.3	87.5	98.3	+11.6
CANDOR _{CYOLO-Mamba}	62.8	65.7	91.5	94.6	99.3	+19.0
CANDOR _{CYOLO} , audio tower as [23]	44.7	47.4	76.2	83.9	96.5	+17.3

Table 2. Components of CANDOR on the real recordings at 0.5 s.

	n	argmax	+transition	full
<i>Perception model</i>				
CYOLO [21]	3	71.3	72.5	91.6 $\pm$ 1.4
CYOLO-SB [22]	6	80.0	86.5	90.6 $\pm$ 0.9
CYOLO-SB+A [†]	3	86.5	90.8	94.4 $\pm$ 0.9
<i>Decision model, on CYOLO-SB candidates (80.0, 86.5)</i>				
residual, no fitted constants	3			88.7 $\pm$ 0.9
without temporal history	3			83.7 $\pm$ 0.4
<i>Image features f_i, on CYOLO-SB candidates</i>				
FILM-conditioned backbone	6			90.6 $\pm$ 0.9
DINOv2 [31]	1			90.9
none	1			90.9

already released one achieves.

Across 48 trained models, accuracy on the synthesized validation split correlates $r = 0.16$ with accuracy on the real recordings (Fig. 2b), and the best-validation model scores 89.3 on real audio, below one reading no image features at all. A reverberant held-out set correlates more strongly ($r = 0.80$) yet still selects poorly. Since the whole field selects models on synthesized audio, this is a caution beyond our own experiments, and it is why our constants are cross-validated on the target recordings instead.

4.2. Score discontinuities

Eq. 3 assumes the performance moves forward, which is what makes the decision tractable and also what a repeat or a jump breaks. Splicing three discontinuities into each performance at random destinations following the protocol of [23], re-running perception on the spliced audio so its conditioning is genuinely wrong afterwards and excluding jumps across a page boundary

where state resets anyway, 41 remain. Over the five seconds after one, CANDOR tracks 8% of onsets against 27% for the argmax it replaces, because the transition model holds the trajectory at a stale position. Triggering a reset on the scorer’s own confidence recovers this to 26% and costs 1.0 point on the unspliced recordings, so selection and discontinuity detection compose at a stated price rather than contend. We report CANDOR without the trigger, the evaluation repertoire being played straight through.

5. CONCLUSION

The synthetic-to-real gap in sheet-image score following has been read as a failure of perception and answered with better audio, more data and new architectures. The measurement reported here places it elsewhere. On real room-microphone recordings the detector already proposes a correct position at every scored onset, and an oracle held to the same realtime commitments recovers almost all of what the deployed rule loses, so at the standard tolerance what fails is the rule that reads a position out of the evidence rather than the evidence itself.

Acting on that, CANDOR orders the candidates the detector already emits and takes real piano recordings from 80.0% to 94.4% of onsets within 0.5 s, against 88.3% for the best published system on a repertoire played straight through, at 1.25 ms per instant on a single CPU thread, without retraining any detector and without layout annotation at test time. The margin holds across three released detectors differing in heads and in training data, and within this framework the decision depends far more on its own temporal history than on the image representations we tested, three of which, including none at all, give the same real-audio accuracy.

Two results temper this. Accuracy on synthesized audio

barely predicts accuracy on real audio across 48 trained models, which makes model selection on a synthetic proxy unreliable for the field and not only for us; and the forward-motion assumption that makes the decision tractable fails at a repeat, where CANDOR is worse than the rule it replaces. Comparing a decoder of CODA’s kind with ours over one shared candidate set is a worthwhile experiment and a different one, and pitch agreement between a candidate and the audio is the natural next source of evidence independent of the history.

Limitations. The correct page is given, all performances are piano, 16 evaluation pieces cannot resolve differences of a point.

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